PSEG Nuclear LLC
P.O. Box 236, Hancocks Bridge, NJ 08038-0236

PSEG
Nuclear LLC

10 CFR 50.73

LR-N23-0064

September 25, 2023

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-001

Hope Creek Generating Station
Renewed Facility Operating License No. NPF-57
Docket No. 50-354

Subject: Licensee Event Report 2023-001-02, Technical Specification Required Shutdown due to Declaring the Suppression Chamber and Primary Containment Inoperable

In accordance with 10 CFR 50.73, PSEG Nuclear LLC submits Licensee Event Report (LER) 2023-001-02. This second supplement removes an erroneous value for the amount of mechanical interference.

There are no regulatory commitments contained in this letter.

Please contact Mr. Harry Balian at (856) 339 - 2173 if you have questions.

Sincerely,

Thomas R. Agster
Plant Manager, Hope Creek Generating Station

Attachment: Licensee Event Report 2023-001-02

cc: USNRC Regional Administrator − Region 1
USNRC NRR Project Manager – Hope Creek
USNRC Senior Resident Inspector – Hope Creek
NJ Department of Environmental Protection, Bureau of Nuclear Engineering

President & Chief Nuclear Officer, PSEG Nuclear LLC
Site Vice President, Hope Creek
Plant Manager, Hope Creek
Vice President, PSEG Nuclear Engineering
Executive Director Regulatory Affairs & Nuclear Oversight
Director Site Regulatory Compliance, Regulatory Affairs
Manager, Nuclear Oversight Corporate
Commitment Coordinator, PSEG Nuclear LLC
Records Management (Record Type 3E.111)

NRC FORM 366 (09-01-2023)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 09/30/2023

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Hope Creek Generating Station	☒ 050 / ☐ 052	354	1 OF 4

4. Title
Technical Specification Required Shutdown due to Declaring the Suppression Chamber and Primary Containment Inoperable

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name		Docket Number
04	30	2023	2023	- 001 -	02	09	25	2023		☐ 050	
									Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		✓ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		✓ 50.73(a)(2)(i)(B)	✓ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Harry Balian	(856) 339 - 2173

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
E	BF	VACB	G202	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
✓ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On April 30, 2023, Hope Creek Generating Station (HCGS) completed a technical specification (TS) required shutdown because limiting condition for operation (LCO) 3.6.1.1, Primary Containment Integrity, was not met. LCO 3.6.1.1 was not met because drywell-to-torus differential pressure could not be established to perform surveillance requirement (SR) 4.6.2.1.f, and LCO 3.6.2.1, SUPPRESSION CHAMBER, could not be met. HCGS demonstrated containment Operability and returned to power operation.

This licensee event report (LER) supplement provides the results of a root cause evaluation (RCE). The direct cause was mechanical interference that prevented seating of one torus-to-drywell vacuum breaker and the root cause was failure to adequately assess and manage risk associated with torus-to-drywell vacuum breaker maintenance.

NARRATIVE

IDENTIFICATION OF OCCURRENCE

Event Date: April 30, 2023
Discovery Date: April 30, 2023

The root cause evaluation (RCE) determined the condition was first identified and misdiagnosed on October 18, 2022. Hope Creek entered an OPERATIONAL CONDITION (OPCON) that was applicable for the limiting conditions for operation (LCO) on October 28, 2022.

CONDITIONS PRIOR TO OCCURRENCE

OPCON 1, POWER OPERATION.

No other structures, systems or components that could have contributed to the event were inoperable during the event.

HCGS was in OPCON 1, POWER OPERATION, when drywell-to-torus bypass leakage was acceptable on September 28, 2022, as shown by successfully completing surveillance requirement (SR) 4.6.2.1.f. HCGS subsequently shutdown to complete a refueling outage. Preventative maintenance was performed on three of the eight torus-to-drywell vacuum breakers during the refueling outage.

HCGS entered OPCON 2 and raised reactor coolant temperature above 200 degrees Fahrenheit on October 28, 2022.

HCGS declared primary containment inoperable while in OPCON 1, POWER OPERATION at 100 percent rated thermal power on April 30, 2023 because Operators could not establish test conditions required to complete SR 4.6.2.1.f.

DESCRIPTION OF OCCURRENCE

On April 30, 2023, HCGS attempted to perform surveillance requirement (SR) 4.6.2.1.f but could not establish the initial test conditions (drywell-to-torus differential pressure). HCGS declared SUPPRESSION CHAMBER and PRIMARY CONTAINMENT INTEGRITY inoperable and completed a plant shutdown in accordance with TS. Event notification (EN) 56495 reported the technical specification required shutdown per 10 CFR 50.72(b)(2)(i) and this LER is required by 10 CFR 50.73(a)(2)(i)(A).

EN 56494 reported a seriously degraded principal safety barrier per 10 CFR 50.72(b)(3)(ii)(A) and a condition that could prevent fulfillment of a safety function per 10 CFR 50.72(b)(3)(v)(D). Subsequent engineering analysis demonstrates that principal safety barriers were not seriously degraded. Therefore, a LER is not required by 10 CFR 50.73(a)(2)(ii). The engineering analysis also demonstrates that the primary containment safety function would be fulfilled if called upon. However, this LER is required by 10 CFR 50.73(a)(2)(v)(D) because primary containment is a single train system that was declared inoperable based on the inability to establish test conditions.

NARRATIVE

DESCRIPTION OF OCCURRENCE (continued)

This LER is also required by 10 CFR 50.73(a)(2)(i)(B) as operation or a condition prohibited by technical specifications (TS) because the RCE established firm evidence that the prohibited condition existed before discovery for a time longer than permitted by TS. The condition was first identified and misdiagnosed on October 18, 2022. On October 28, 2022, HCGS entered OPCON 2 and then OPCON 1. LCOs 3.6.1.1 and 3.6.2.1 are applicable in OPCON 2 and OPCON 1. Consequently, HCGS operated in a condition prohibited by TS for roughly six months. Further, HCGS inadvertently violated LCO 3.0.4 when entering OPCON 2 and 1, because LCO 3.0.4 prohibits ascent into an applicable OPCON when an LCO is not met.

CAUSE OF EVENT

HCGS determined the direct cause was torus-to-drywell vacuum breaker 'A' was not set appropriately to prevent mechanical interference from its associated Limit Switches or Closing Magnet Assemblies. This prevented the vacuum breaker from fully seating.

HCGS determined the root cause was failure to assess a first-call preventive maintenance (PM) on torus-to-drywell vacuum breakers as high risk. Consequently, the outage work management processes did not establish readiness milestones, appropriate readiness challenges, and effective risk management actions.

SAFETY CONSEQUENCES AND IMPLICATIONS

There were no safety consequences because of this event. An engineering analysis determined that primary containment and the suppression chamber (torus) were capable of performing their intended safety functions. Therefore, per NEI 99-02, “Regulatory Assessment Performance Indicator Guideline,” Revision 7, this will not be counted as a safety system functional failure.

PREVIOUS EVENTS

No similar LER or corrective action occurrences were found following a review of station LERs and the corrective action program for the past three years.

LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Hope Creek Generating Station	■ 050 / ☐ 052	354	2023	001	02

NARRATIVE

CORRECTIVE ACTIONS

HCGS identified bypass leakage through one torus-to-drywell vacuum breaker, completed repairs, satisfactorily performed SR 4.6.2.1.f, and returned to power operation on May 6, 2023.

HCGS established corrective actions to improve procedures so that high risk and non-recoverable work is better defined and identified. Non-recoverable and infrequently performed work associated with TS structures, systems, or components will be defined as high risk. Risk classification for identified non-recoverable activities will be reassessed. An extent of condition review for similar work will be completed. Risk classification of all eight torus-to-drywell vacuum breaker PMs will be reassessed.

HCGS established additional corrective actions to resolve the contributing causes. These include procedure enhancements, reinforcing station expectations, and fleet wide communications of the learnings provided by this event.

COMMITMENTS

There are no regulatory commitments in this LER.